A Calibrated Placebo Test for Time-Varying Features in Credit Risk Models

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Abstract

Credit-risk models increasingly use time-varying inputs such as macroeconomic indicators and text-based sentiment. The usual validation check is to add the feature and test whether discrimination (AUC) improves. This paper shows that this check can be fooled: when the feature is a persistent time series and the model has no control for the credit cycle, a feature that is pure noise passes the check 38% of the time in our baseline simulation and 67% of the time when the cycle drifts (nominal 5%). A placebo test that generates fake series with the same persistence as the real one restores the intended size (2% to 5% in every setting tested). Placebo series that are simply shuffled are valid only when period effects are controlled: without them they pass 33% to 43% of useless series. With year effects in the model, the standard check is close to correctly sized under moderate cycles (7%) but over-rejects when the cycle drifts (12%), while the persistence-matched placebo remains at 2%. The price of this protection is power: with a persistent feature and a moderate true effect, detection falls from 73% to 30% (effect 0.2). Two public loan datasets illustrate the point: a tone index passes the standard check on Lending Club with year effects (AUC gain 0.0056, 95% CI 0.0027 to 0.0088) but not the persistence-matched placebo ($p = 0.71$). We give a one-page validator checklist.

Keywords: credit risk, IFRS 9, model validation, placebo test, time-varying features, spurious regression, text sentiment

JEL: G21, C52, C12

1. Introduction

Banks and supervisors expect a credit model to be validated before use. Under IFRS 9 the expected-credit-loss model must reflect forward-looking information, and the ECB guide on internal models expects validators to challenge the explanatory variables. A natural way to add forward-looking information is a time-varying feature: a macro indicator, a market index, or a sentiment measure built from central-bank or news text.

The validator's question is simple: does the feature really help? The common answer is to add it and compare AUC on a test set, often with a bootstrap confidence interval. If the interval excludes zero, the feature is accepted.

This paper shows why that answer can be wrong. A time-varying feature takes one value per period, so thousands of loans share the same value. If that series is persistent (it drifts slowly, as most macro and text series do) it will, by chance, line up with the unobserved credit cycle in the sample. A bootstrap over loans cannot see this, because it treats the loans as the only source of randomness. The feature then looks useful when it is not.

What the literature has and has not done. Spurious regression with persistent series is classical (Granger and Newbold, 1974; Phillips, 1986) and the age-period-cohort identification problem in credit scoring is well documented (Breeden et al., 2008; Breeden and Leonova, 2023). Recent machine-learning work feeds time-varying and local economic information into default models: Barbaglia et al. (2023) use local economic conditions in European mortgages, Breeden and Leonova (2023) feed age-period-cohort functions into ML scores as fixed offsets, and Kündig and Sigrist (2025) add a latent space-time process to boosted trees. We did not find a paper that combines the two: a calibrated, simulation-tested validation test for time-varying features in a loan-level classifier, with explicit size and power results. This is a literature check by search, not a proof that none exists, and we invite correction.

Contribution. The paper makes five contributions:

1. We show in simulation where the standard check fails and where it does not.
2. We show that shuffled placebos are valid only with period controls, and give a persistence-matched placebo that is valid in all settings tested.
3. We measure the power cost of the stricter test.
4. We illustrate the tests on two public loan datasets.
5. We give a short checklist for validators.

Placebo, permutation and surrogate-series tests are general statistical tools; we do not claim them as new. This paper builds on Nawaz (2026), which applied a placebo test to validating narrative features in credit models. The present paper extends that application. The earlier headline results used period controls, where shuffled placebos remain valid and the standard check rarely passed noise; the uncontrolled pass rates reported there are consistent with Figure 1 below. This paper adds the persistence-matched placebo, which is needed whenever period controls are absent or imperfect, and measures size and power by simulation.

2. The problem and the proposed test

Setting. Loan *i* in period *t* defaults with probability given by a logistic model of loan characteristics, an unobserved credit cycle *c(t)* and, possibly, a true effect of a time series *s(t)*. The validator holds a baseline model and a candidate model that adds *s(t)* as one extra column. Both are fitted on a training sample and compared by AUC on a test sample. The question is whether the AUC gain is more than chance.

Why the standard check can fail. The standard check resamples test loans and asks how uncertain the gain is. It treats *s(t)* as fixed. But *s(t)* has only as many independent values as there are periods, and if it is persistent it has far fewer. If the credit cycle is also persistent, any *s(t)* can happen to move with it. The gain is then real in the sample but is not evidence about *s(t)*, and the loan-level bootstrap does not reveal it.

The placebo test. We ask a different question: how large a gain would a series with no true effect produce? The steps are:

1. Fit the baseline model and the candidate model with the real series; record the AUC gain.
2. Generate K = 99 placebo series that are unrelated to default but resemble the real series.
3. For each placebo, refit the candidate model and record its gain. These gains are the empirical null.
4. The p-value is (1 + number of placebos with gain at least as large as the real gain) / (K + 1).

The key choice is how placebos are generated. We compare two:

- **Shuffled (iid):** randomly reorder the real series across periods. This destroys persistence.
- **Persistence-matched:** draw an AR(1) series with the same first-order autocorrelation as the real series (estimated from it, floored at zero).

Period controls. We also vary whether the models include period (year) effects. When they do, the series enters as its deviation from the within-year mean, so only within-year variation is tested. This removes the slow component but also removes some information, which is the price we measure in Section 4.

3. Simulation design

We simulate 40 quarters (10 years) with 600 loans per quarter, 24,000 loans in all. Each loan has five standard-normal characteristics with fixed coefficients (0.8, -0.6, 0.5, 0.4, -0.3) and an intercept of -3. The unobserved credit cycle is an AR(1) process with persistence 0.9 and standard deviation 0.5 (the "moderate cycle"), or persistence 0.99 and standard deviation 1.0 (the "drifting cycle"). The candidate series is an independent standardised AR(1) with persistence 0, 0.5 or 0.9, and enters the true model with coefficient gamma of 0, 0.1, 0.2 or 0.3. Gamma = 0 is the null: the series is pure noise.

Loans are split at random, 70% to training and 30% to testing. Each cell is run for 300 replications. Each replication runs the standard check (100 bootstrap draws over the test set, pass if the lower 2.5th percentile of the gain exceeds zero) and the two placebo tests (99 placebos each, pass if p is below 0.05). We report pass rates with 95% Wilson intervals. With 300 replications, a rate near 5% has an interval of roughly 3% to 8%, so small differences are noise.

Table 1. Simulation settings. Moderate cycle: persistence 0.9, standard deviation 0.5. Drifting cycle: persistence 0.99, standard deviation 1.0.

Setting	Period control	Cycle	Series persistence	Role
A	Year effects	Moderate	0	Easy case
B	Year effects	Moderate	0.5, 0.9	Persistent series, controlled
C	Year effects	Drifting	0 and 0.9	Hard cycle
D	None	Moderate	0 and 0.9	Uncontrolled
E	None	Drifting	0.9	Uncontrolled, hard cycle

Table 1 lists the settings. The simulation is deliberately simple (a logistic scorecard, a cycle independent of the loan features). Section 7 discusses what that does and does not show.

4. Simulation results

Without a period control, the standard check is badly fooled, and so are shuffled placebos. When the series is pure noise but persistent (0.9), the standard check passes it 38% of the time under a moderate cycle and 67% under a drifting cycle (Figure 1, right). Shuffled placebos pass it 33% and 43%. Persistence-matched placebos pass it 5% and 5%, the intended rate.

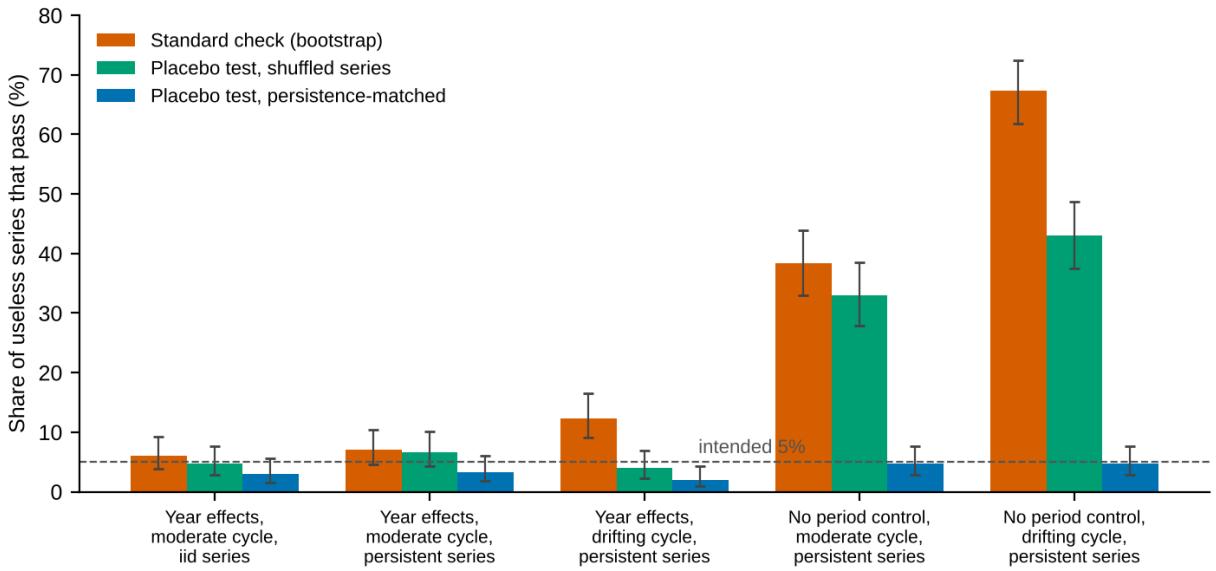


Figure 1. Share of useless series (true effect = 0) that pass each test. Bars show 95% Wilson intervals; 300 replications per setting.

With year effects, the problem shrinks but does not vanish. For persistent noise, the standard check passes 7% under a moderate cycle (consistent with 5%) and 12% under a drifting cycle (interval 9% to 17%, so above 5%). Shuffled placebos pass 7% and 4%. Persistence-matched placebos pass 3% and 2%, slightly conservative.

The cost is power. Figure 2 shows how often a real effect is detected. With an independent, non-persistent series, the two placebo tests detect an effect of 0.2 in 72% to 73% of runs, against 49% for the standard check. With a persistent series, detection is much lower for everyone: at 0.2 it is 30% for the persistence-matched placebo, 37% for shuffled and 26% for the standard check. The shuffled placebo looks more powerful here only because it is too lenient on persistent series; it is not a fair comparison once persistence is present.

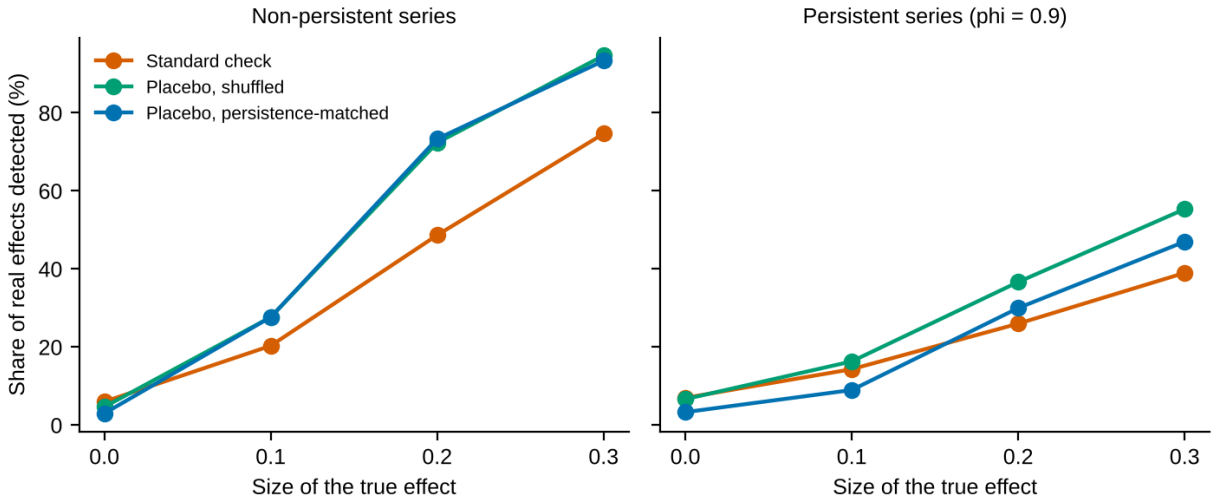


Figure 2. Share of real effects detected, with year effects and a moderate cycle. 300 replications per point.

Table 2. Share of runs in which each test passes the series, by true effect size. Persistent series (0.9), moderate cycle, 300 replications per row. Effect 0 is the null, so the figure is the false-pass rate.

Setting	Standard check	Shuffled placebo	Persistence-matched placebo
No period control, effect 0	38%	33%	5%
Year effects, effect 0	7%	7%	3%
Year effects, effect 0.1	14%	16%	9%
Year effects, effect 0.2	26%	37%	30%
Year effects, effect 0.3	39%	55%	47%

Table 2 summarises these rates. Two points deserve plain statement:

1 The persistence-matched test is the only one with acceptable size in every setting, but it is not free: when the series is persistent and the effect is small to moderate, it will often miss a real effect.

2 No test here recovers power that the data do not contain. A persistent series gives few independent observations, and no method can change that.

5. Real-data illustration

We apply the tests to two public datasets, as an illustration and not as a general finding.

- **Lending Club (US).** Loans matched to the most recent FOMC statement before issue. The series is a net-sentiment score of 75 statements (persistence 0.45).
- **Bondora (Estonia and other EU markets).** Loans matched to the most recent ECB statement. The series is a net-sentiment score of 165 statements (persistence 0.73).

In each case the baseline is a gradient-boosted classifier on loan characteristics, trained on a random 60,000-loan subsample with a 25% test split, and the candidate model adds the sentiment score. We use no period control, year effects, or quarter effects (the series then enters as its deviation from the within-period mean). We run 99 placebos of each type. For Lending Club, default means a resolved loan that was charged off or in default (fully paid loans are the non-defaults). For Bondora, it means a default within 12 months of origination, using only loans with a full 12-month window. Neither is the Basel 90-days-past-due definition.

Table 3. Real-data results. AUC gain of the candidate model over the baseline on the test sample; 60,000-loan subsamples; 99 placebos of each type. Standard check passes if the bootstrap interval lies above zero. p-values are empirical.

Dataset	Period control	AUC gain	95% bootstrap interval	Standard check	p, shuffled	p, persistence-matched
Lending Club	None	0.0020	-0.0003 to 0.0046	Fail	0.86	0.94
Lending Club	Year	0.0056	0.0027 to 0.0088	**Pass**	0.61	0.71
Lending Club	Quarter	-0.0010	-0.0028 to 0.0013	Fail	0.13	0.17
Bondora	None	0.0043	0.0018 to 0.0072	**Pass**	0.01	0.04
Bondora	Year	0.0000	-0.0017 to 0.0026	Fail	0.90	0.90
Bondora	Quarter	0.0006	-0.0010 to 0.0024	Fail	0.40	0.47

Table 3 reports the results. **Lending Club with year effects is the cautionary case.** The AUC gain is positive and its interval excludes zero, so the standard check accepts the feature. But a persistence-matched placebo produces gains this large or larger in 71% of draws. The mean placebo gain (0.0061) exceeds the real gain (0.0056). The apparent improvement is what an arbitrary slow series of this kind delivers.

Bondora with no period control is a borderline case. The standard check passes, the shuffled placebo gives p = 0.01, and the persistence-matched placebo gives p = 0.04. The shuffled test is the more extreme because it ignores persistence. Once year effects are added the gain disappears (0.0000). This is consistent with the sentiment series proxying for slow time variation that year effects already capture. With one run and 99 placebos, p = 0.04 is not strong evidence either way.

What this does not show. These results do not say sentiment is useless in credit risk, and they do not show it works. They show that on these two samples, with these models, the AUC gain is not distinguishable from what a slow noise series would produce once the series is treated as the persistent time series it is. Results rely on one 60,000-loan subsample per dataset and a single random split, and both datasets are consumer credit.

6. Practical guidance for validators

1. **Treat a time-varying feature as having as many observations as it has distinct periods**, not as many as there are loans. Report that number (75 statements, 40 quarters) next to every claim about the feature.
2. **Always include period effects** in both baseline and candidate models, and test the feature as a within-period deviation. This was the single largest protection in our simulations (the standard check's false-pass rate fell from 38% to 7%).
3. **Run a persistence-matched placebo test** with at least 99 placebos. Estimate the series' first-order autocorrelation, draw series with that persistence, and report the empirical p-value alongside the bootstrap interval.
4. **Do not use shuffled placebos** on persistent series unless period effects are in the model; without them they passed 33% to 43% of useless series.
5. **Report power.** A non-significant result on a persistent feature may mean the data cannot tell. State the smallest effect the test could have detected.
6. **Treat a pass on the standard check but a fail on the placebo test as a red flag**, and document it in the validation report.

The cost of these steps is small: a placebo run refits the candidate model about 100 times, which is practical for the scorecard-sized models we tested.

7. Limitations

- **The simulation is simple.** The baseline is a logistic scorecard, the cycle is independent of loan features, and the series is independent of the cycle under the null. Real series (policy tone, macro indicators) may be correlated with the true cycle, in which case a "useless" series is not independent noise. The test as designed asks whether the series adds anything beyond what AR(1) noise of the same persistence would, which is a specific null and not the only one.

- **AR(1) matching is one choice.** Real series can have trends, breaks or longer memory. Block bootstrap or phase-randomised placebos are alternatives we have not tested.
- **Random train-test split.** We split loans at random, not out of time. An out-of-time design is closer to validation practice and may change the gains.
- **Few replications.** 300 per setting gives intervals of about plus or minus 3 to 5 points near 5%, so differences of a few points are not informative. We also use K = 99 placebos, so the smallest possible p-value is 0.01.
- **Real data are an illustration.** One 60,000-loan subsample per dataset, one gradient-boosting configuration, one random split, and non-Basel default definitions. The Bondora borderline result (p = 0.04) is not strong evidence.
- **Year and quarter effects are not the only credit-cycle controls.** Vintage effects, macro covariates and age-period-cohort models (Breeden and Leonova, 2023) are alternatives and may interact with the test.
- **No claim about regulatory acceptance.** The checklist is a proposal, not a supervisory expectation.

8. Conclusion

A feature that varies only over time carries far less information than its loan-level sample size suggests. In simulation, the common check of an AUC gain with a bootstrap interval accepted a useless persistent series in 38% to 67% of runs when no period control was used, and 7% to 12% with year effects. A placebo test with persistence-matched series kept the false-pass rate at 2% to 5% in every setting, at a cost in power. On two public datasets the same logic overturned an apparent gain (Lending Club with year effects). The practical message for validators is short: control for period, count periods not loans, and compare the gain with what a persistent noise series would give.

Replication. Code for the simulation, the real-data runs and the figures accompanies this paper (see the replication package).

Disclosure of AI use

Claude (Anthropic), a generative AI assistant, was used for language correction, for searching relevant literature, and for drafting and correcting the Python code. The author ran and checked the code and the results, and takes full responsibility for the content. No AI system is listed as an author.

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